Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 3

GROUND STATE ENERGIES

E0 (Ha) : -230.75351694
E1 (Ha) : -230.57821527
Gap (Ha) : 0.17530166
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,5)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 5
Spin-orbitals   : 10
Hilbert dim   : 100

SPIN PROPERTIES

<S2>   : -0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 2.961507
<D> / site : 0.592301

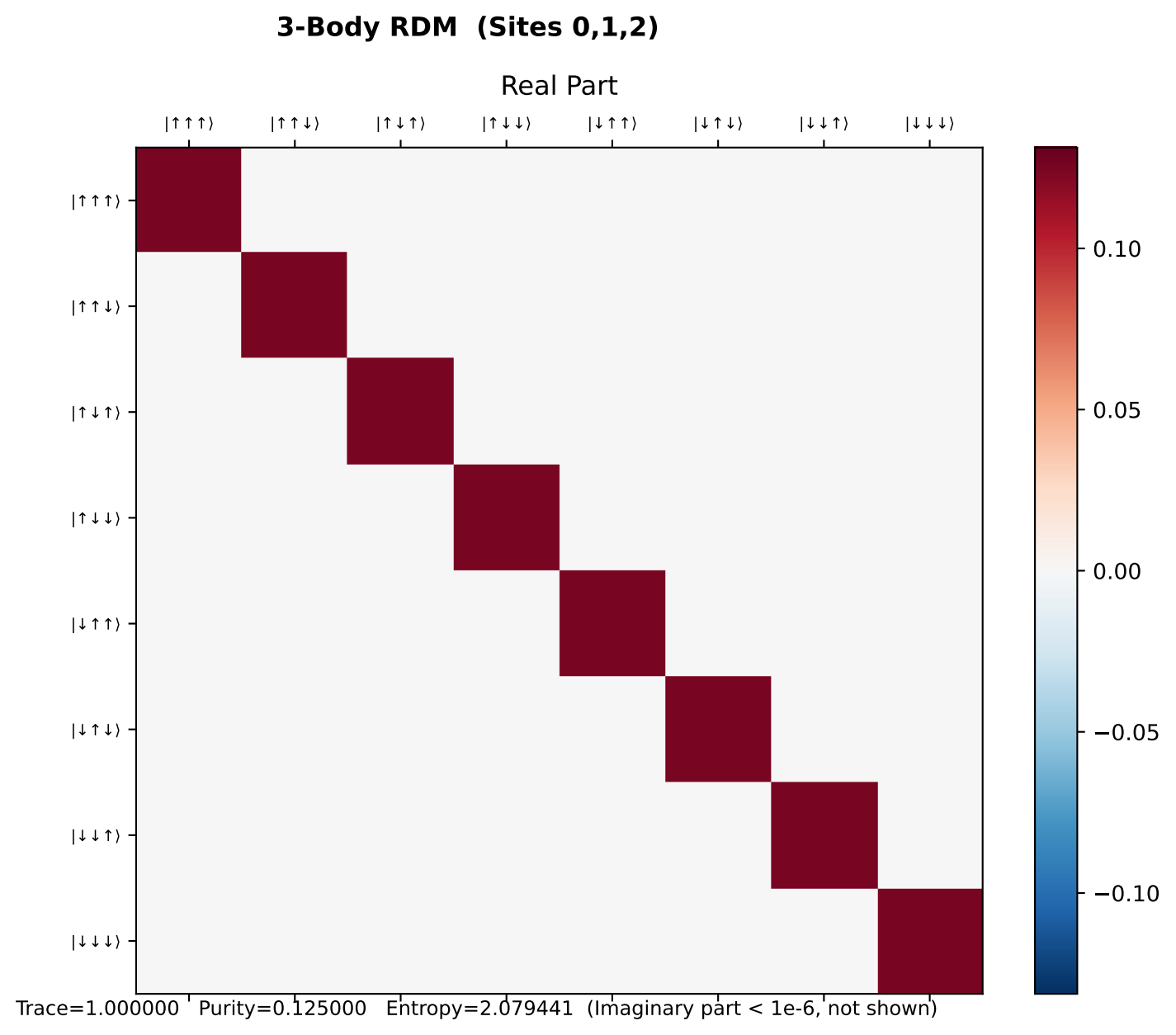
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Body order : 3
Clusters   : 10

Date: 2026-07-06 18:31

Summary — 3-Body RDMs (Benzene, CAS(6,5))

Orbitals	Purity	C_GME (pure)	C_LB	C_gcg	C_stiefel	C_UB	State
0,1,2	0.125000	—	0.000000	0.000319	—	0.000000	STRONGLY MIXED
0,1,3	0.148891	—	0.000000	0.002603	—	0.000000	STRONGLY MIXED
0,1,4	0.125559	—	0.000000	0.001190	—	0.000000	STRONGLY MIXED
0,2,3	0.125559	—	0.000000	0.000337	—	0.000000	STRONGLY MIXED
0,2,4	0.148904	—	0.000000	0.000525	—	0.000000	STRONGLY MIXED
0,3,4	0.125949	—	0.000000	0.000001	—	0.000000	STRONGLY MIXED
1,2,3	0.496207	—	0.000000	0.028891	—	0.088963	MIXED
1,2,4	0.486887	—	0.000000	0.002873	—	0.088517	MIXED
1,3,4	0.499669	—	0.000000	0.070681	—	0.087795	MIXED
2,3,4	0.499782	—	0.000000	0.073909	—	0.087901	MIXED


3-Body RDM (Sites 0,1,2)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125000 Entropy=2.079441 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,1,2)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1250000348
  Von Neumann entropy  : 2.0794414025
  Eigenvalue min       : 1.248858e-01
  Eigenvalue max       : 1.250381e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12500003

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

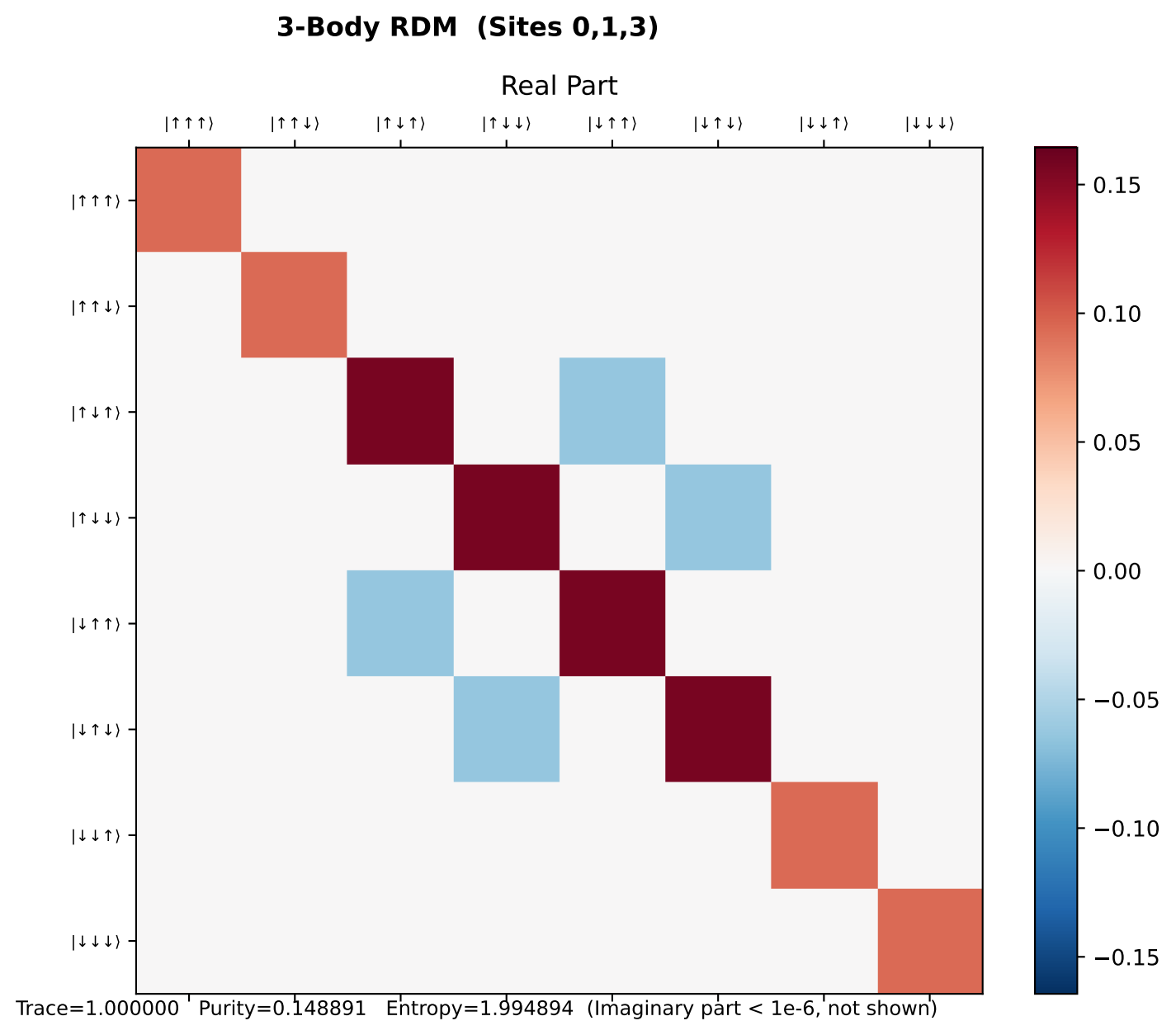
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.00000001

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.105606  C_GME=0.000015
2. p=0.068840  C_GME=0.000056
3. p=0.117924  C_GME=0.000135
4. p=0.115502  C_GME=0.001845
5. p=0.091682  C_GME=0.000017
... and 5 more states
```


3-Body RDM (Sites 0,1,3)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.148891 Entropy=1.994894 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,1,3)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1488911426
  Von Neumann entropy  : 1.9948940631
  Eigenvalue min       : 9.344902e-02
  Eigenvalue max       : 2.196529e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.14889114

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

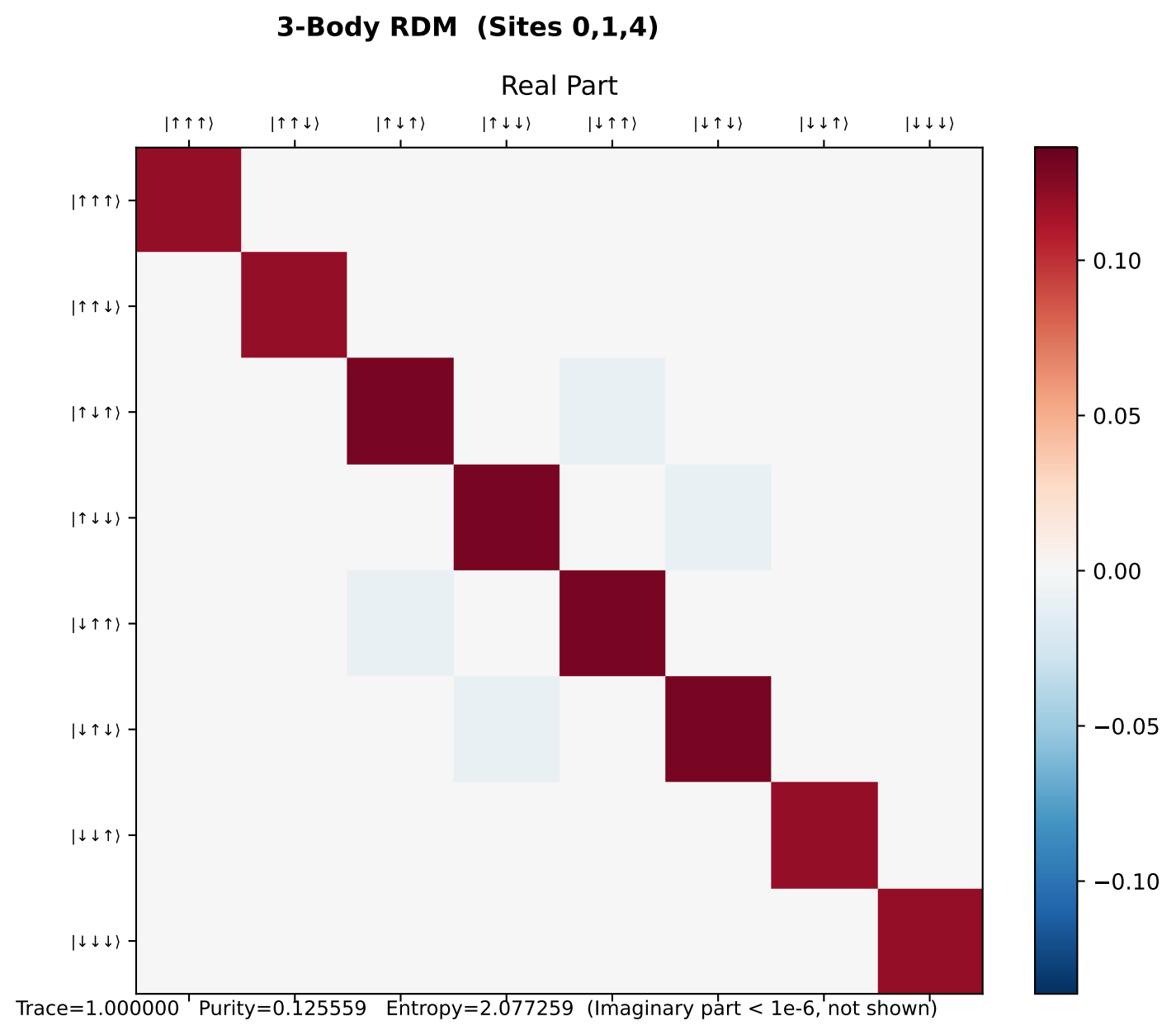
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.00000000

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.102787  C_GME=0.000000
2. p=0.099340  C_GME=0.000002
3. p=0.084777  C_GME=0.000000
4. p=0.130842  C_GME=0.013100
5. p=0.088127  C_GME=0.009901
... and 5 more states
```


3-Body RDM (Sites 0,1,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125559 Entropy=2.077259 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,1,4)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1255591892
  Von Neumann entropy  : 2.0772587151
  Eigenvalue min       : 1.201730e-01
  Eigenvalue max       : 1.394809e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12555919

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

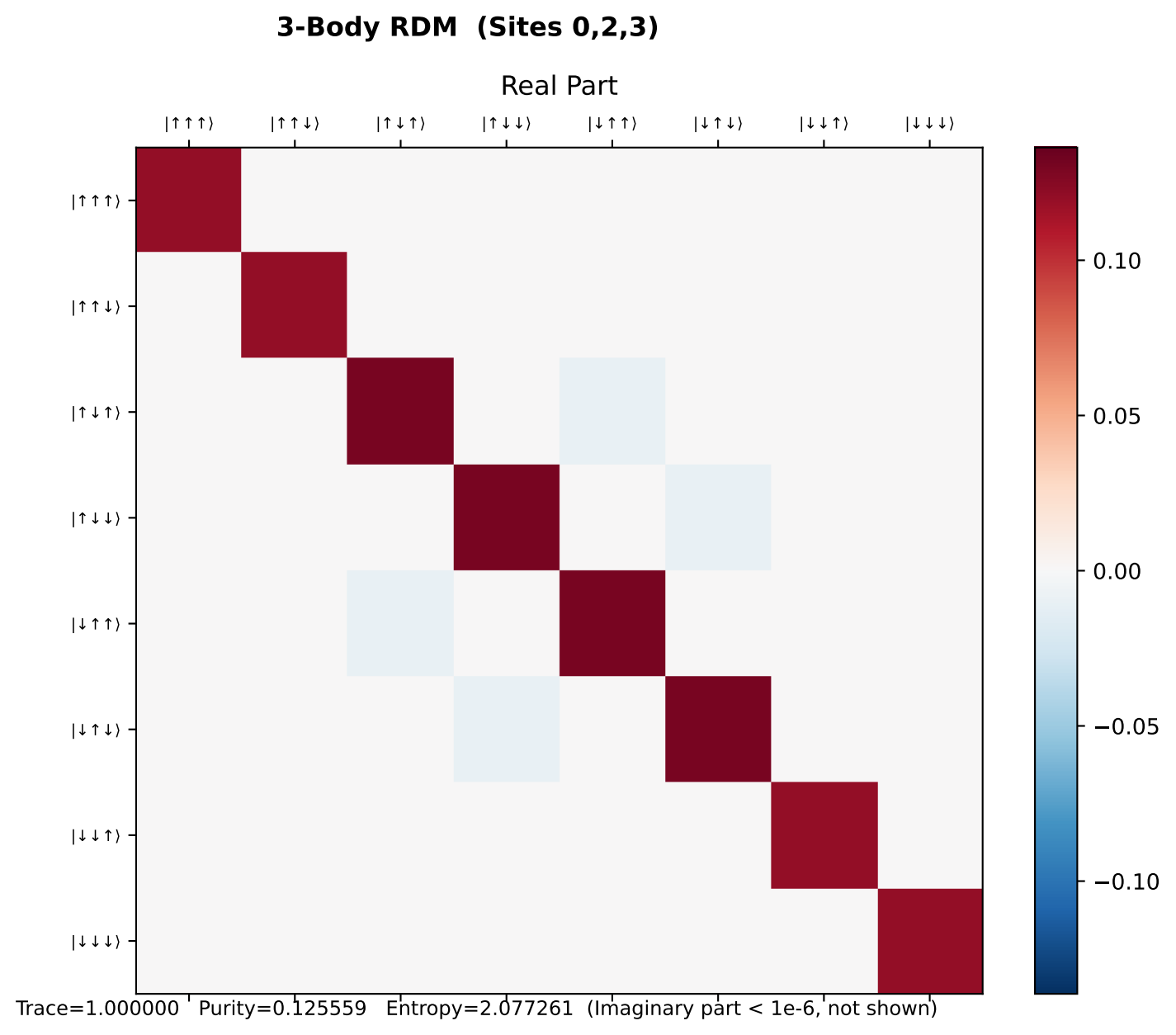
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.00000001

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.119853  C_GME=0.003399
2. p=0.125250  C_GME=0.000021
3. p=0.082407  C_GME=0.000046
4. p=0.104214  C_GME=0.001017
5. p=0.105438  C_GME=0.000000
... and 5 more states
```


3-Body RDM (Sites 0,2,3)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125559 Entropy=2.077261 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,2,3)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1255585209
  Von Neumann entropy  : 2.0772612935
  Eigenvalue min       : 1.201759e-01
  Eigenvalue max       : 1.394722e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12555852

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

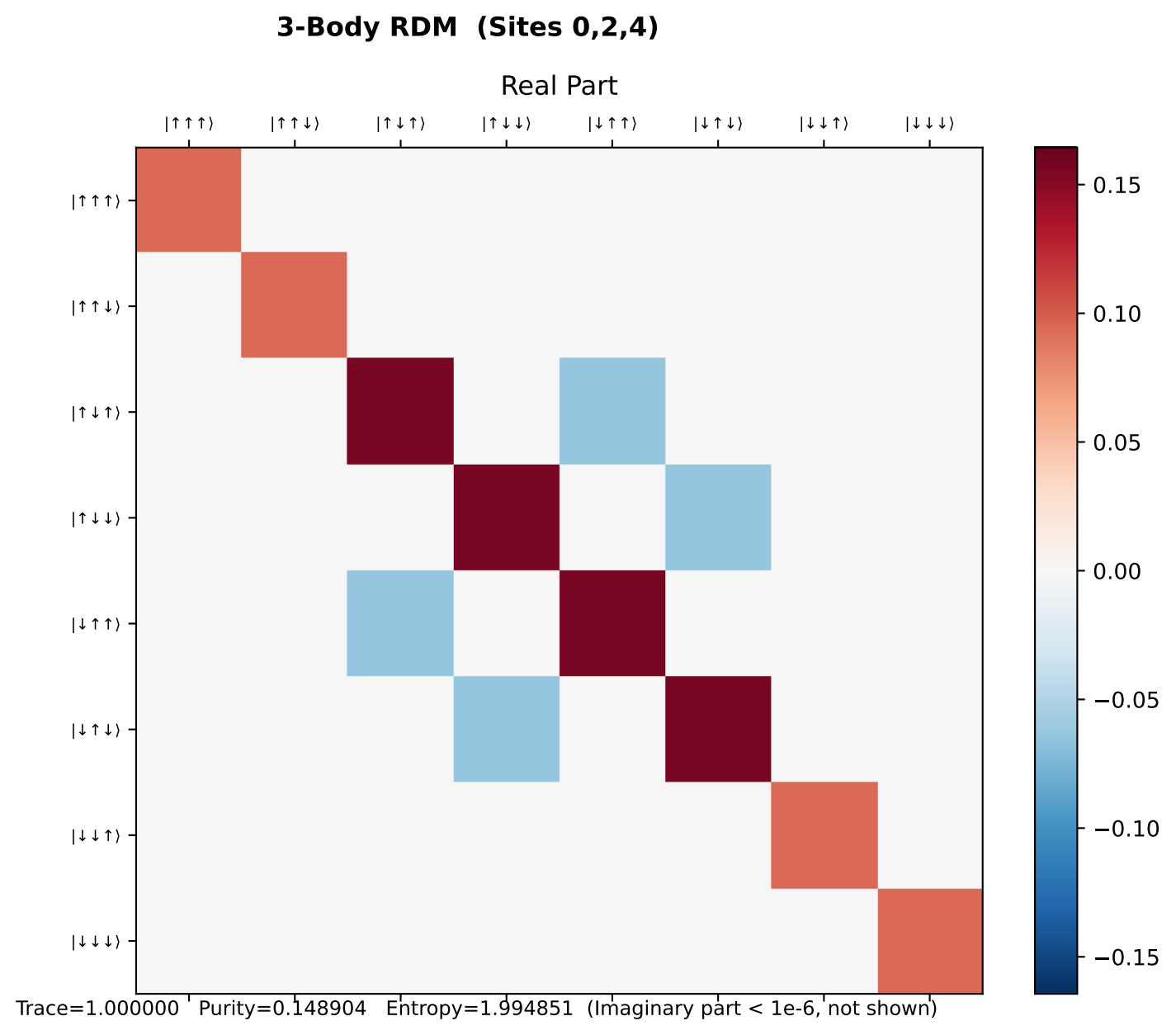
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.00000001

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.104433  C_GME=0.000000
2. p=0.085273  C_GME=0.000006
3. p=0.100016  C_GME=0.000000
4. p=0.119682  C_GME=0.000000
5. p=0.109507  C_GME=0.000000
... and 5 more states
```


3-Body RDM (Sites 0,2,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.148904 Entropy=1.994851 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,2,4)

RDM Metrics

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.1489040060
Von Neumann entropy	1.9948505094
Eigenvalue min	9.344053e-02
Eigenvalue max	2.196784e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis

gamma = Tr(rho^2) = 0.14890401

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

Convex Roof Analysis

Lower Bounds:
- C_LB = 0.00000000

C_GME (unitary optimization):
- C_stiefel (unitary opt) = nan BEST

C_GME (unitary optimization - Expoenetial Map):

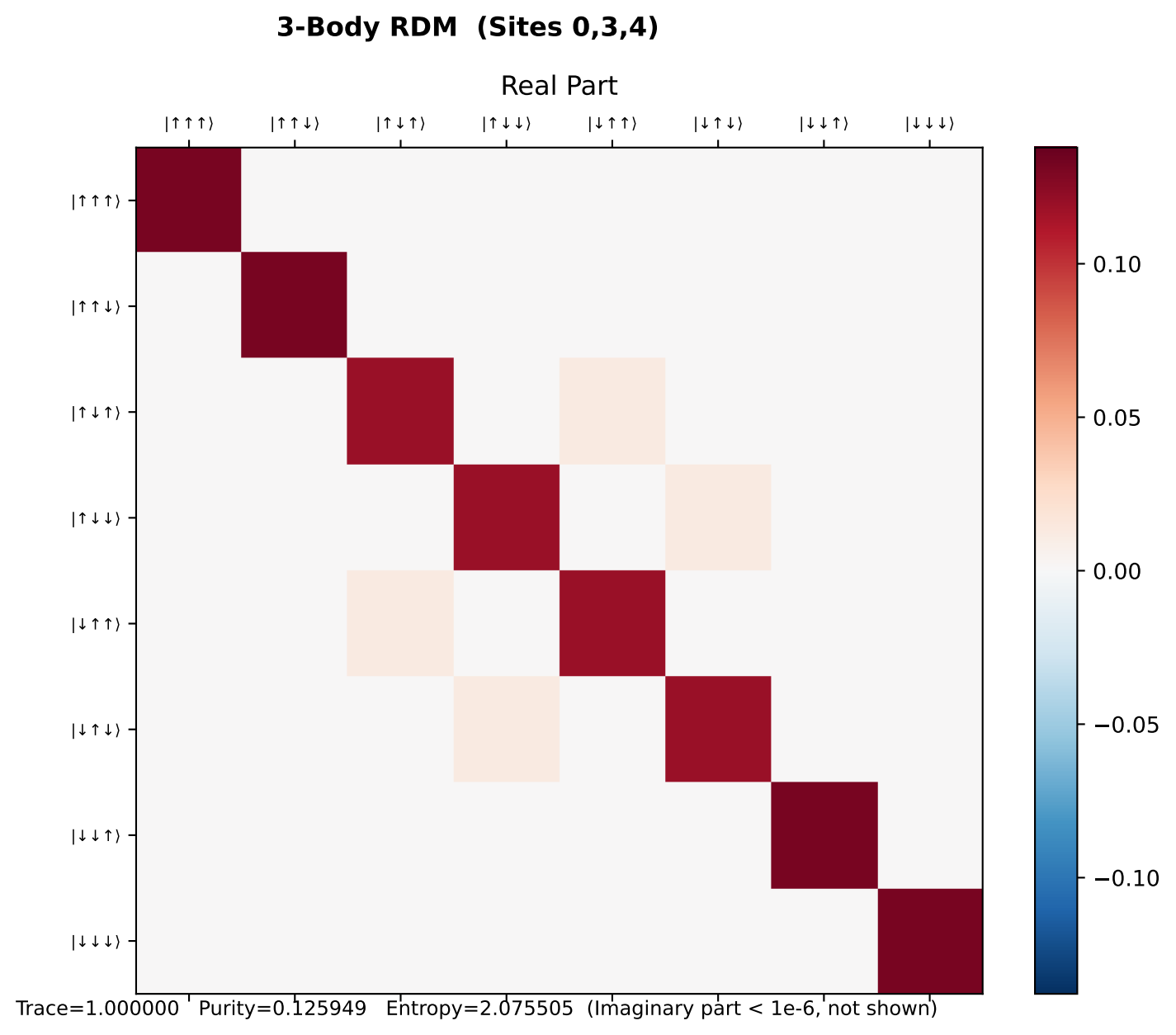
Upper Bound:
- C_UB (eigendecomp) = 0.00000001

Status: inconclusive

Solver source:
- –

1. p=0.097317 C_GME=0.000000
2. p=0.108378 C_GME=0.000000
3. p=0.081953 C_GME=0.000000
4. p=0.104202 C_GME=0.000000
5. p=0.074520 C_GME=0.001232

... and 5 more states


3-Body RDM (Sites 0,3,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125949 Entropy=2.075505 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,3,4)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1259492576
  Von Neumann entropy  : 2.0755048607
  Eigenvalue min       : 1.061328e-01
  Eigenvalue max       : 1.312891e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12594926

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

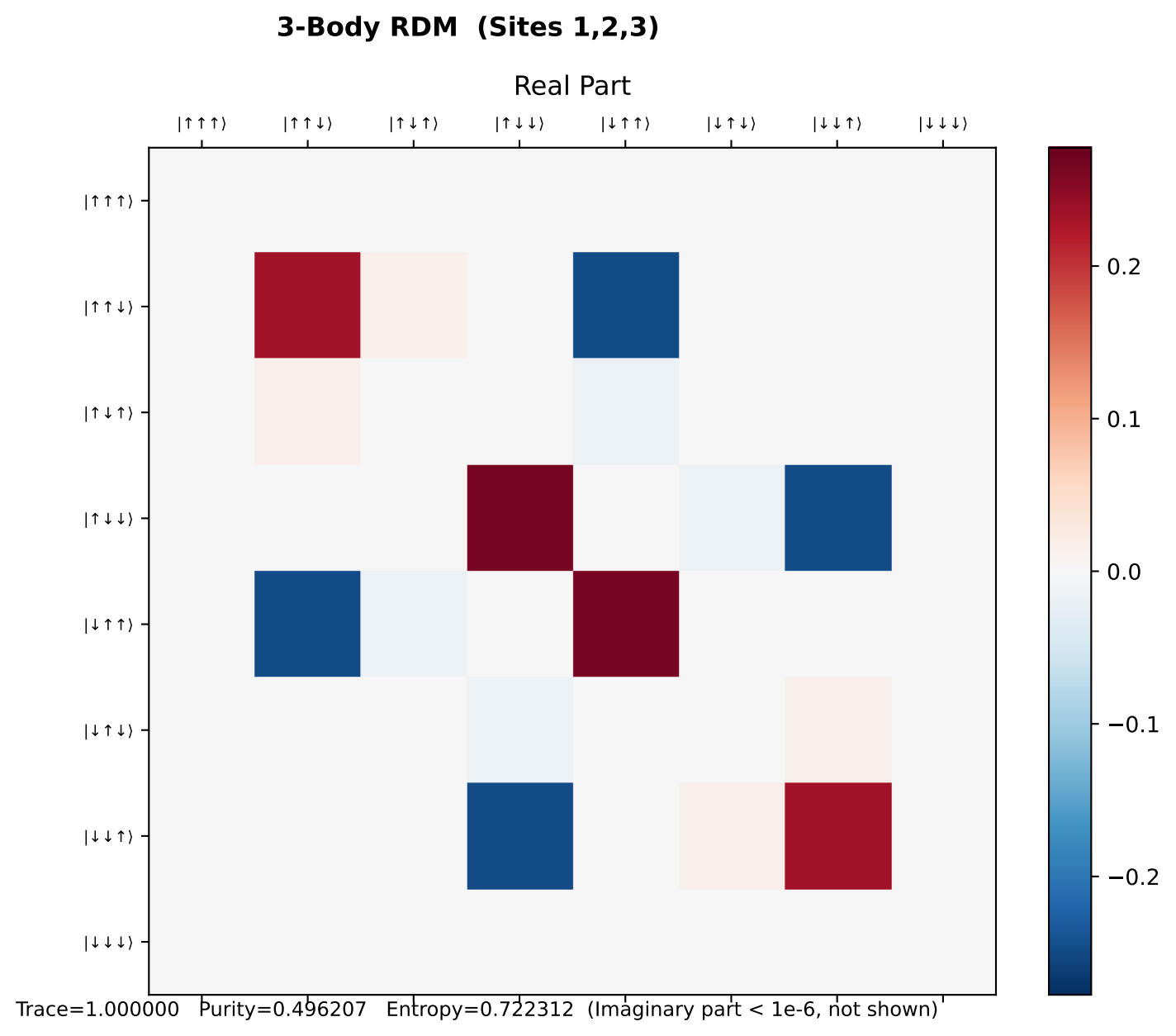
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.00000001

  Status: no_GME

  Solver source:
    –
```

```
1. p=0.078294  C_GME=0.000000
2. p=0.109383  C_GME=0.000000
3. p=0.099869  C_GME=0.000000
4. p=0.096236  C_GME=0.000000
5. p=0.097404  C_GME=0.000000
... and 5 more states
```


3-Body RDM (Sites 1,2,3)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.496207 Entropy=0.722312 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 1,2,3)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4962065490
  Von Neumann entropy  : 0.7223117676
  Eigenvalue min       : 6.338489e-04
  Eigenvalue max       : 4.980985e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.49620655

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-----------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

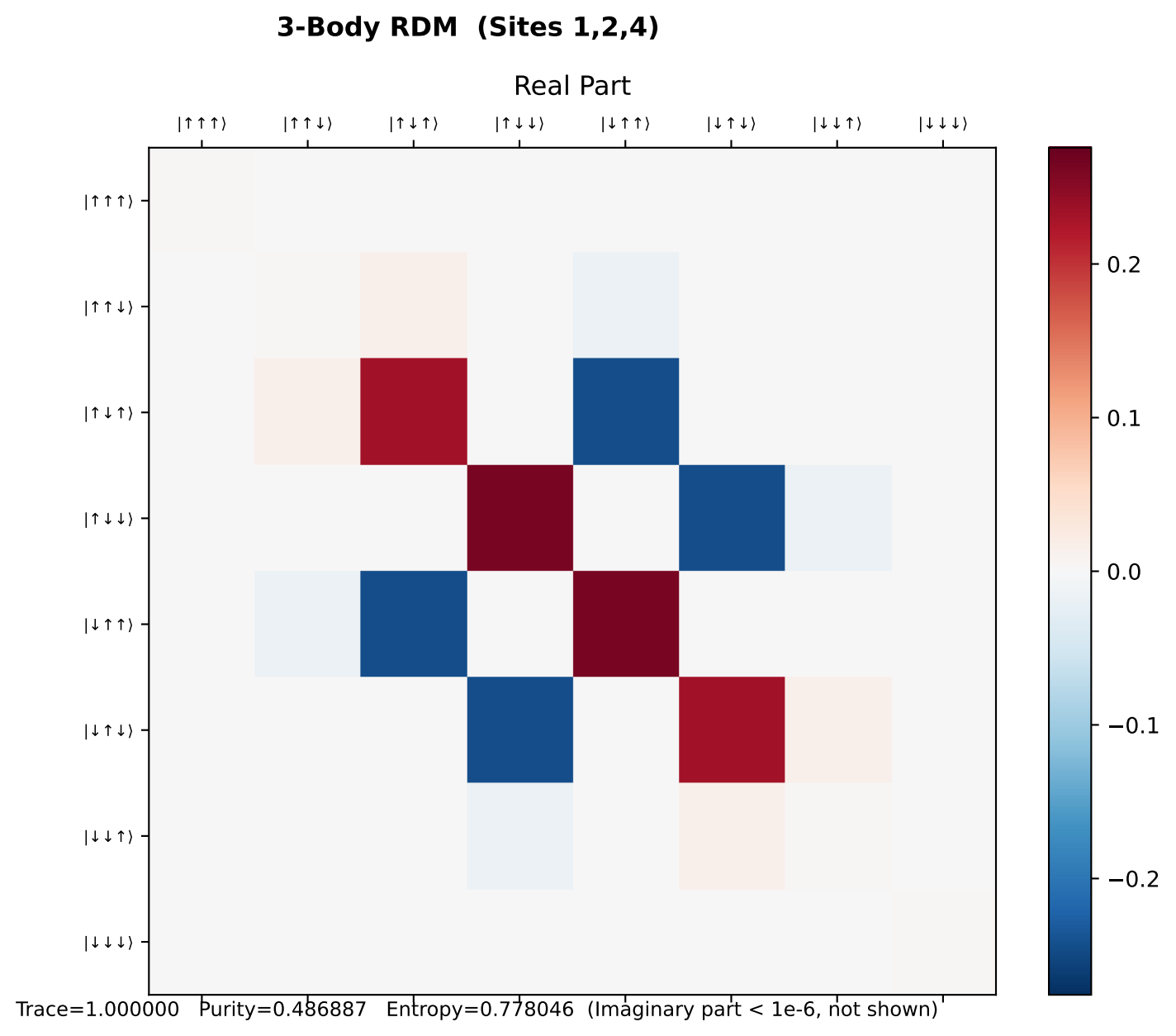
```
Convex Roof Analysis
===================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.08896268

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.014067  C_GME=0.358505
2. p=0.157274  C_GME=0.027355
3. p=0.162682  C_GME=0.031762
4. p=0.023549  C_GME=0.019195
5. p=0.135504  C_GME=0.023572
... and 5 more states
```


3-Body RDM (Sites 1,2,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.486887 Entropy=0.778046 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 1,2,4)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4868869169
  Von Neumann entropy  : 0.7780459582
  Eigenvalue min       : 2.204961e-03
  Eigenvalue max       : 4.933851e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.48688692

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

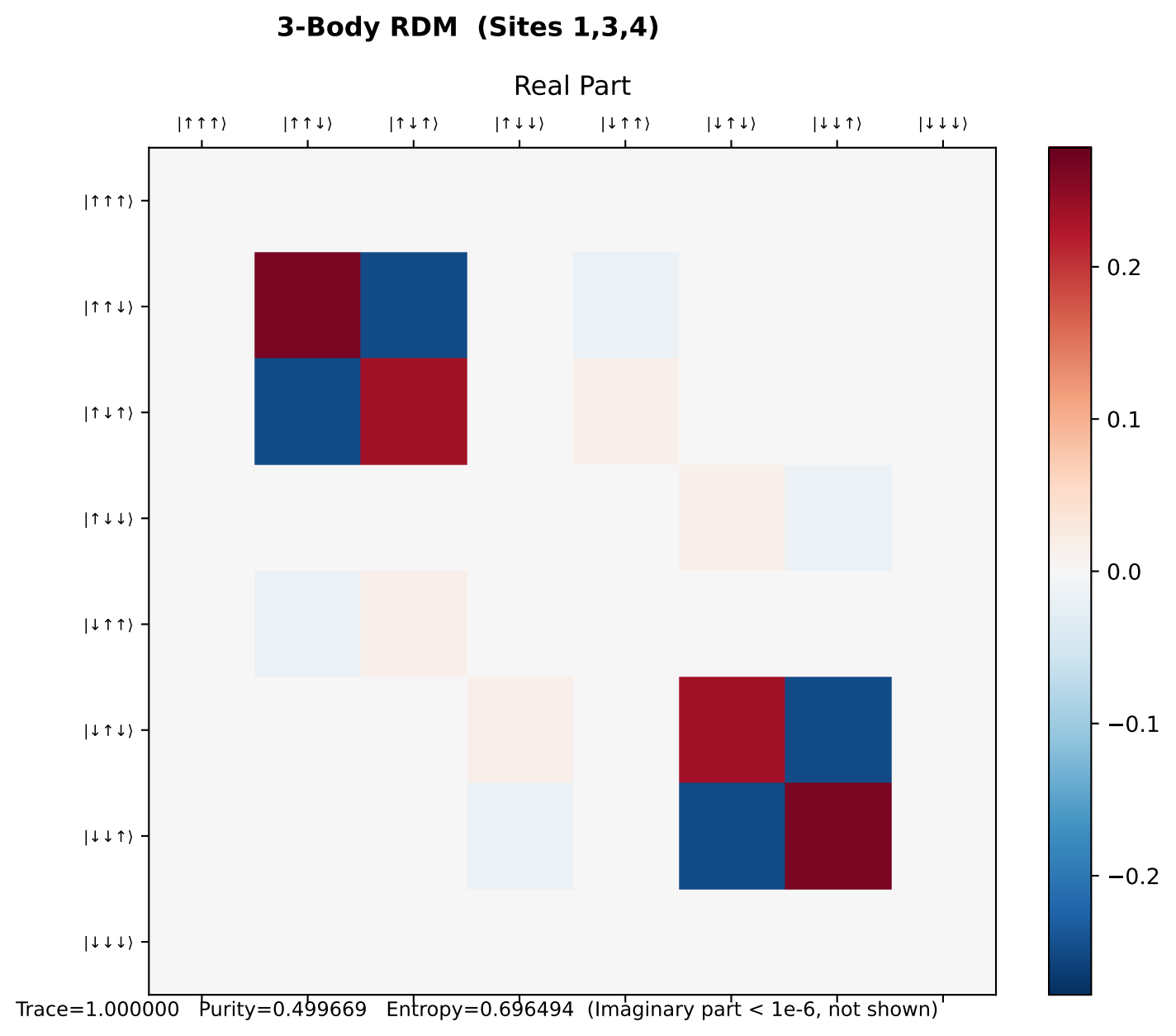
```
Convex Roof Analysis
===================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.08851678

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.162555  C_GME=0.000000
2. p=0.131407  C_GME=0.000000
3. p=0.150347  C_GME=0.000000
4. p=0.117094  C_GME=0.000017
5. p=0.221171  C_GME=0.000000
... and 5 more states
```


3-Body RDM (Sites 1,3,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.499669 Entropy=0.696494 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 1,3,4)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4996691458
  Von Neumann entropy  : 0.6964935362
  Eigenvalue min       : 5.515454e-05
  Eigenvalue max       : 4.998345e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.49966915

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

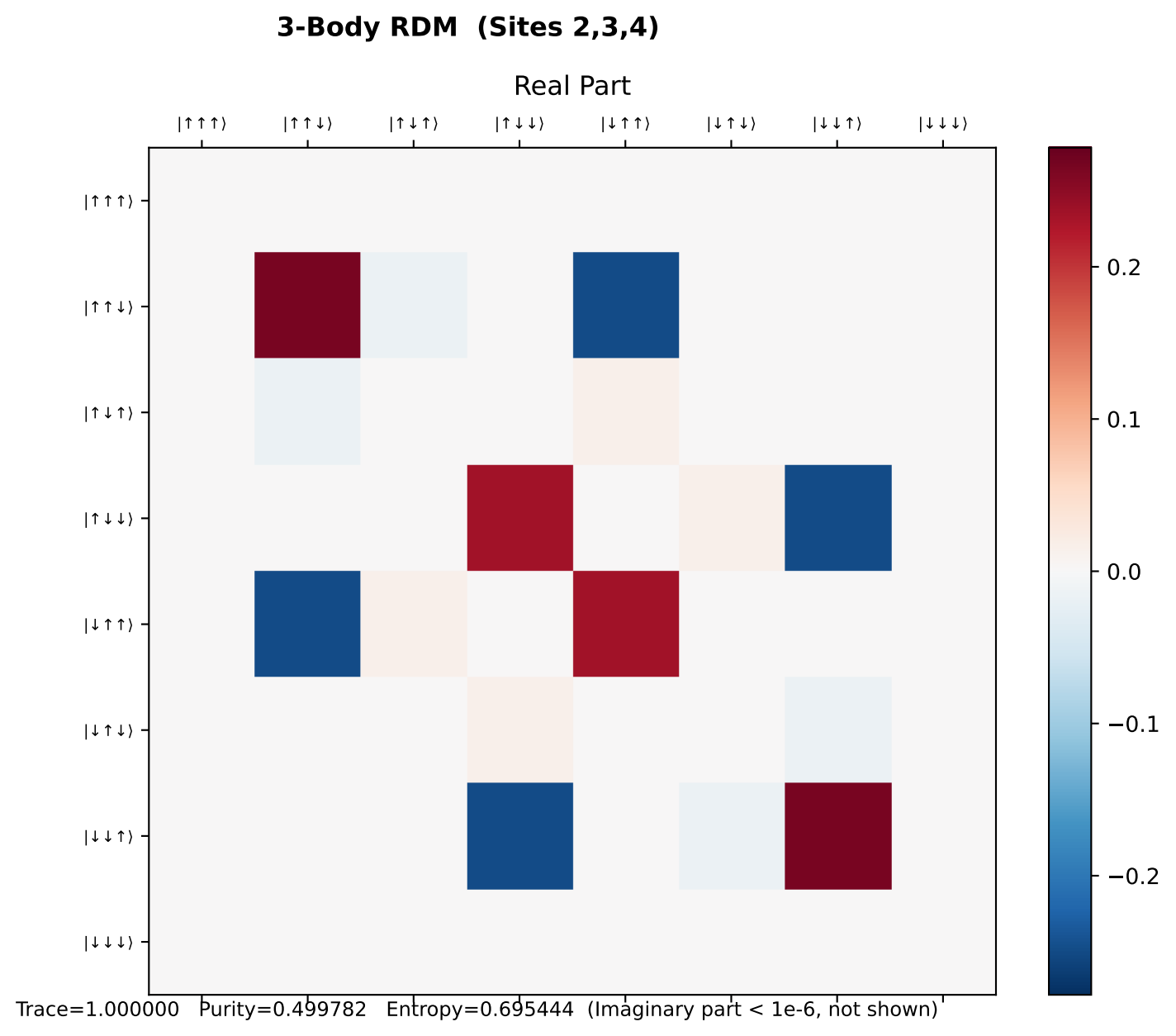
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.08779489

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.031647  C_GME=0.010789
2. p=0.166680  C_GME=0.071063
3. p=0.137054  C_GME=0.071769
4. p=0.137248  C_GME=0.082931
5. p=0.160812  C_GME=0.075980
... and 4 more states
```


3-Body RDM (Sites 2,3,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.499782 Entropy=0.695444 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 2,3,4)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.4997819190
Von Neumann entropy	0.6954436893
Eigenvalue min	3.635211e-05
Eigenvalue max	4.998909e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.49978192

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

Convex Roof Analysis
==========

Lower Bounds:
C_LB = 0.00000000
C_GME (unitary optimization):
C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.08790080

Status: inconclusive

Solver source:
–

1. p=0.077701 C_GME=0.077348
2. p=0.069934 C_GME=0.045648
3. p=0.098935 C_GME=0.055351
4. p=0.163311 C_GME=0.070531
5. p=0.076923 C_GME=0.099885
... and 4 more states